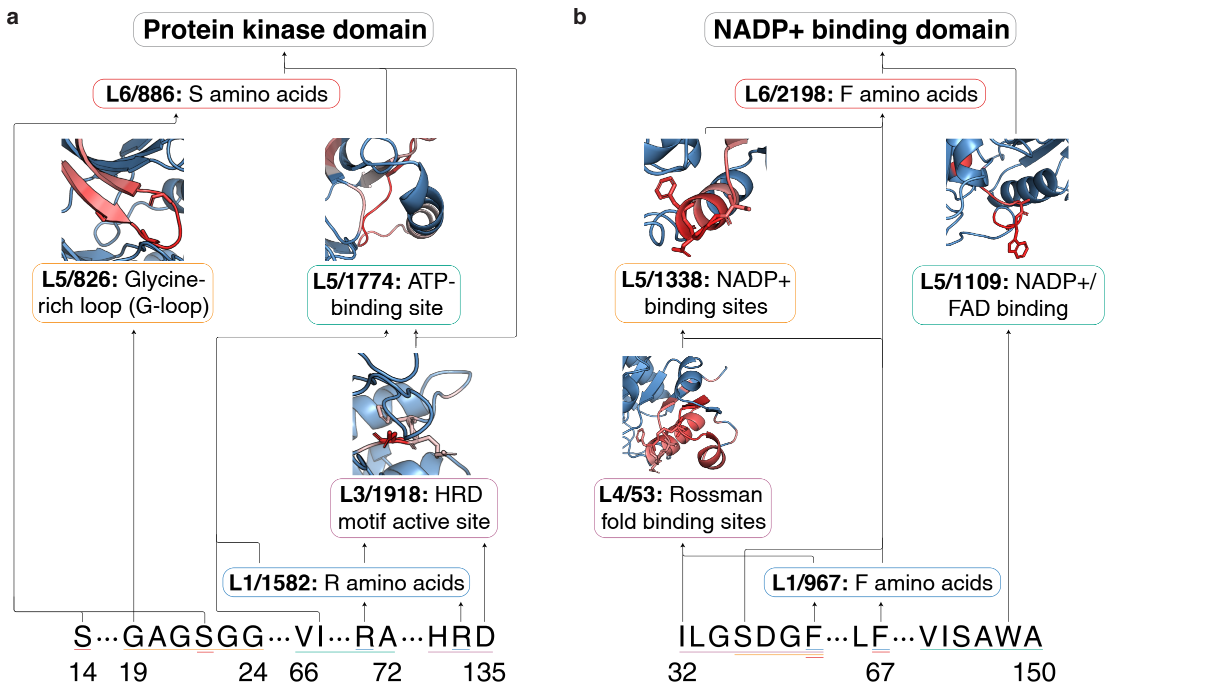

Figure 3. **Examples of family circuits discovered using ProtoMech.** We use the ProtoMech visualization tool to examine **a**, protein kinase domain and **b**, NADP+ binding domain circuits. We find interpretable features related to binding and active sites, secondary structure, and biochemical patterns. We observe that earlier layers are detecting key amino acids that assemble into complex motifs.

stabilizes the protein and binds to Fc (Sauer-Eriksson et al., 1995). This feeds into layer 3 (L3/3027), which detects hydrophobic interactions, most notably between leucine (L) at position 5 and W43. The interaction between L5 and W43 is well-documented in literature, and is known to contribute to GB1's high stability (Gronenborn et al., 1991). Separately, L1/249 also feeds into layer 6 (L6/1610), which similarly detects W amino acids, mirroring our findings from the NADP+ binding domain with ESM2 learning repeating concepts.

We then leveraged the DMS assay from (Olson et al., 2014) to investigate the mechanistic drivers of high-fitness variants. To this end, we analyzed a known high-fitness variant, A24W (alanine to tryptophan at position 24) (Fig. 4b). Interestingly, while all the latents that made up the circuit from the wildtype remain active, ProtoMech identifies a feature previously not seen in the wildtype, L6/3029, which activates on inward-facing and hydrophobic residues. This suggests that A24W, by introducing a bulky aromatic amino acid, packs more efficiently into the protein's hydrophobic core, improving stability and therefore binding affinity.

Conversely, we also analyzed a known low-fitness variant, W43Q (Tryptophan to Glutamine at position 43) (Fig. 4c). Given that W43 is a critical interface residue, we hypothesized that the circuit would reflect a loss of both binding affinity and stability. Indeed, feeding the mutated sequence into ProtoMech reveals the deactivation of the W amino acid detectors L1/249 and L6/1610, confirming the loss of a binding site. Crucially, the mutation also weakens L6/3027, which previously activated hydrophobic interactions. This suggests that W43Q destabilizes GB1 and destroys the Fc-binding site, thereby preventing GB1 from performing its function. These examples underscore ProtoMech's ability to detect high- and low-fitness mutations and to provide interpretable, biologically plausible explanations.

4. Discussion

Contrasting Sparse Autoencoders. ProtoMech fundamentally differs from SAE-based approaches to interpreting pLMs. While SAEs focus on decomposing a single activation state (see Appendix G for more details), ProtoMech functions as a replacement model by explicitly learning the computation going from one representation to the next. Although recent work has utilized pretrained SAEs to infer circuit connectivity in pLMs (Nainani et al., 2025), such methods are still reliant on decomposing single activation states and thus cannot serve as a replacement model.

Denoising ESM2. Surprisingly, we observe that our discovered circuits occasionally *outperform* the performance of the full ESM2 model on both protein family classification and function prediction. This phenomenon is most pronounced when ESM2 struggles. For instance, in protein family classification tasks where ESM2 achieves an F1 score below 0.5

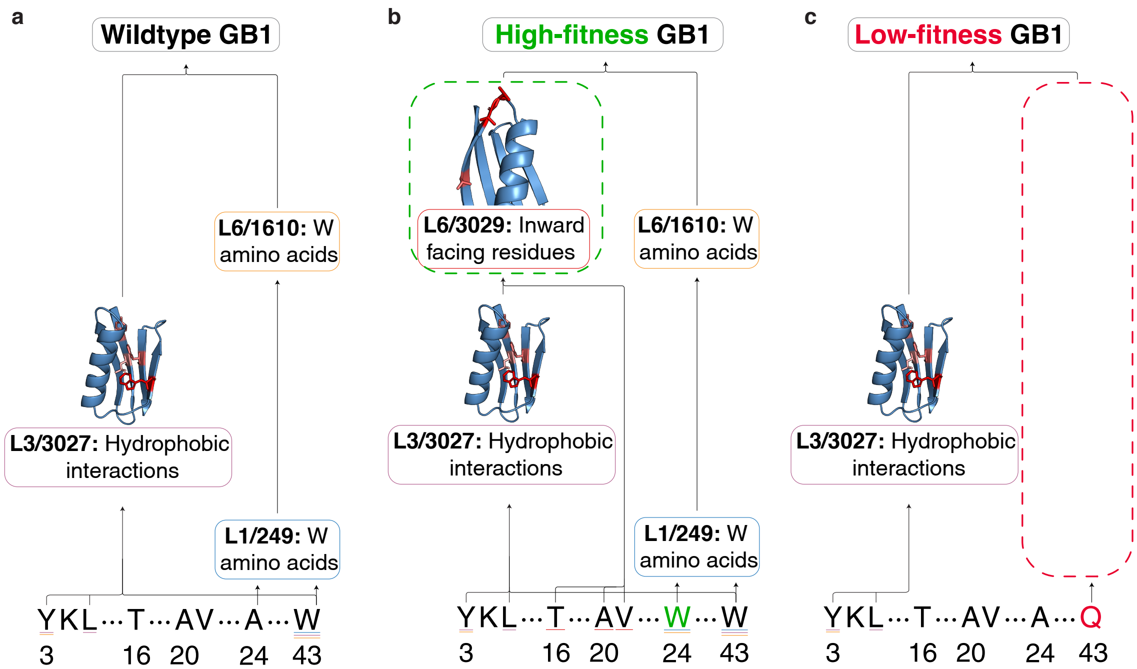

Figure 4. **Examples of the mutation sensitivity of function circuits discovered using ProtoMech. a**, We feed in the wildtype sequence of GB1 into ProtoMech and identify latents related to binding affinity and stability. **b**, A high-fitness variant of GB1 activates an additional latent that corresponds to protein stability. **c**, Conversely, a low-fitness variant of GB1 deactivates parts of the circuit related to binding affinity and stability. These findings highlight ProtoMech's ability to provide a mechanistic rationale for changes in fitness to sequences.

(0.39 ± 0.04), ProtoMech circuits achieve a higher average performance (0.43 ± 0.27) (Fig. 8). In fitness prediction tasks, although ESM2 retains a higher Spearman correlation on average, 37% of our circuits achieve a higher performance. We attribute this to a *denoising* (regularization) effect inherent to ProtoMech's sparse latent representation. By transcoding ESM2 through a sparse latent space, ProtoMech effectively filters out task-irrelevant noise, distilling the computation down to ESM2's most salient features. This suggests that ProtoMech offers utility beyond interpretability, potentially as a regularizer to enhance pLM robustness.

High-Throughput Protein Screening. Across our case studies, we demonstrate that ProtoMech captures known biological motifs and distinguishes between high- and low-fitness mutations. One potential application of ProtoMech in this regard is high-throughput protein screening, which necessitates generating and screening thousands of sequences to create large-scale functional libraries (Wu et al., 2019; Walton et al., 2025b). In this context, ProtoMech offers a solution as a mechanistic filter. By tracing computational pathways, researchers can better identify variants utilizing biologically-plausible motifs, enabling the selection of higher-quality candidates for wet-lab synthesis.

Limitations. One limitation of ProtoMech is the number of parameters required to train a CLT. Due to the CLT's dependence on cross-layer connections, the number of decoder matrices required scales with $\mathcal{O}(L^2)$, compared to $\mathcal{O}(L)$ with the number of encoder matrices. In practice, training a CLT for the ESM2-8M model required approximately 28M parameters, a $3.5\times$ increase over the original model's size. However, this upfront computational cost is effectively amortized in circuit discovery, as we do not need to retrain the CLT. One interesting future direction includes scaling up ProtoMech to larger variants of ESM2, possibly by reducing the computational burden of CLTs.

Another limitation lies in the interpretation of ProtoMech circuits, which currently relies on manual analysis coupled with existing biological annotations. As our capacity to label circuits is bounded by the current state of biological knowledge, it is possible that there exists circuits governing mechanisms that are not yet well-characterized. Furthermore, our reliance on human interpretation to label our circuits restricts the scale of our qualitative analysis. Developing automated annotation pipelines remains a critical next step to accelerate the discovery of novel biological insights.

Conclusion. In this paper, we introduce ProtoMech, a framework for tracing computational circuits in protein language models via cross-layer transcoders. We demonstrate that these circuits are highly compressible and steerable toward high-fitness sequences, while consistently overlapping with known biological motifs. Our work opens the door toward principled circuit tracing in pLMs.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning, specifically toward interpreting protein language models. The potential societal impacts of our work are overwhelmingly positive. Potential applications of this work include discovering biological mechanisms, enhancing protein design, and improving the performance of downstream tasks.

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References

Abramson, J., Adler, J., Dunger, J., Evans, R., Green, T., Pritzel, A., Ronneberger, O., Willmore, L., Ballard, A. J., Bambrick, J., Bodenstein, S. W., Evans, D. A., Hung, C.-C., O'Neill, M., Reiman, D., Tunyasuvunakool, K., Wu, Z., Žemgulytė, A., Arvaniti, E., Beattie, C., Bertolli, O., Bridgland, A., Cherepanov, A., Congreve, M., Cowen-Rivers, A. I., Cowie, A., Figurnov, M., Fuchs, F. B., Gladman, H., Jain, R., Khan, Y. A., Low, C. M. R., Perlin, K., Potapenko, A., Savy, P., Singh, S., Stecula, A., Thillaisundaram, A., Tong, C., Yakneen, S., Zhong, E. D., Zielinski, M., Žídek, A., Bapst, V., Kohli, P., Jaderberg, M., Hassabis, D., and Jumper, J. M. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 630(8016):493–500, May 2024.

Adams, E., Bai, L., Lee, M., Yu, Y., and AlQuraishi, M. From mechanistic interpretability to mechanistic biology: Training, evaluating, and interpreting sparse autoencoders on protein language models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267, pp. 460–476, 2025.

Ameisen, E., Lindsey, J., Pearce, A., Gurnee, W., Turner, N. L., Chen, B., Citro, C., Abrahams, D., Carter, S., Hosmer, B., Marcus, J., Sklar, M., Templeton, A., Bricken, T., McDougall, C., Cunningham, H., Henighan, T., Jermyn, A., Jones, A., Persic, A., Qi, Z., Ben Thompson, T., Zimmerman, S., Rivoire, K., Conerly, T., Olah, C., and Batson, J. Circuit tracing: Revealing computational graphs in language models. Transformer Circuits Thread, 2025. URL `https://transformer-circuits.pub/2025/attribution-graphs/methods.html`.

Araya, C. L., Fowler, D. M., Chen, W., Muniez, I., Kelly, J. W., and Fields, S. A fundamental protein property, thermodynamic stability, revealed solely from large-scale measurements of protein function. *Proceedings of the National Academy of Sciences*, 109(42):16858–16863, 2012.

Bhatnagar, A., Jain, S., Beazer, J., Curran, S. C., Hoffnagle, A. M., Ching, K. S., Martyn, M., Nayfach, S., Ruffolo, J. A., and Madani, A. Scaling unlocks broader generation and deeper functional understanding of proteins. In *Advances in Neural Information Processing Systems*, 2025.

Boeckmann, B., Bairoch, A., Apweiler, R., Blatter, M.-C., Estreicher, A., Gasteiger, E., Martin, M. J., Michoud, K., O'Donovan, C., Phan, I., Pilbout, S., and Schneider, M. The SWISS-PROT protein knowledgebase and its supplement TrEMBL in 2003. *Nucleic Acids Research*, 31(1):365–370, 2003.

Bossemeyer, D. The glycine-rich sequence of protein kinases: A multifunctional element. *Trends in Biochemical Sciences*, 19(5):201–205, 1994.

Corominas, G. B., Stocco, F., and Ferruz, N. Sparse autoencoders in protein engineering campaigns: Steering and model diffing. In *ICML 2025 Generative AI and Biology (GenBio) Workshop*, 2025.

Ding, D., Shaw, A. Y., Sinai, S., Rollins, N., Prywes, N., Savage, D. F., Laub, M. T., and Marks, D. S. Protein design using structure-based residue preferences. *Nature Communications*, 15(1):1639, 2024.

Dunefsky, J., Chlenski, P., and Nanda, N. Transcoders find interpretable LLM feature circuits. In *Advances in Neural Information Processing Systems*, volume 37, pp. 24375–24410, 2024.

Elhage, N., Nanda, N., Olsson, C., Henighan, T., Joseph, N., Mann, B., Askell, A., Bai, Y., Chen, A., Conerly, T., DasSarma, N., Drain, D., Ganguli, D., Hatfield-Dodds, Z., Hernandez, D., Jones, A., Kernion, J., Lovitt, L., Ndousse, K., Amodei, D., Brown, T., Clark, J., Kaplan, J., McCandlish, S., and Olah, C. A mathematical framework for transformer circuits. Transformer Circuits Thread, 2021. URL `https://transformer-circuits.pub/2021/framework/index.html`.

Faure, A. J., Domingo, J., Schmiedel, J. M., Hidalgo-Carcedo, C., Diss, G., and Lehner, B. Mapping the energetic and allosteric landscapes of protein binding domains. *Nature*, 604(7904):175–183, 2022.

Gallagher, T., Alexander, P., Bryan, P., and Gilliland, G. L. Two crystal structures of the B1 immunoglobulin-binding